GovFix: Campus Workflow Engine

Problem Understanding, Solution Proposal & System Architecture Report

VORTEXA 3.0 Hackathon
Domain 7: Government Systems & GovTech
Selected Track: **University Service Requests**
Team: **OrbitX** | Oct 11, 2026

1. Problem Understanding

Higher education institutions—such as **Dr. D. Y. Patil Institute of Technology, Pimpri**—operate as high-density academic campuses housing thousands of students, faculty, and research personnel. Every semester generates thousands of operational service requests spanning:

- **Campus Facilities & Infrastructure**: Laboratory power line trips, classroom smart-board/projector failures, hostel water pump outages.
- **Academic IT & Computing Systems**: GPU workstation driver freezes in Computer Labs, departmental Wi-Fi AP failures, server link drops.
- **Administrative & Equipment Requests**: Classroom allocation shifts, asset transfers, and equipment calibration clearances.

Despite campus ERP portals (e.g., Samarth eGov, basic Google Forms, or standard ticketing tools), request lifecycles systematically collapse due to **five structural failure modes**:

1. The "Limbo" Problem
Tickets enter "In Progress" and sit for weeks without technician progression forcing or SLA accountability.

2. Premature & Ghost Closures
Technicians mark tickets "Closed" without physical repair photos, supervisor sign-off, or student confirmation.

3. Silent SLA Breaches
Deadlines expire unnoticed because legacy databases lack daemons to auto-escalate cases to HODs or the Registrar.

4. The Flaky AI Trap
Unconstrained AI chatbots hallucinate status jumps, approve invalid closures, or drop records when given mutation rights.

> **Critical Failure: Relational database rows in legacy ERPs are mutable. If a laboratory hazard is neglected, traditional databases cannot cryptographically prove when deadlines were missed or if historical records were retroactively modified.**

2. Proposed Solution: GovFix (University Track)

GovFix is an institutional-grade, backend-enforced finite-state workflow engine purpose-built for university campus maintenance and IT operations, governed by one ironclad statutory principle:

> **"Deterministic Rules Govern Authority, State Transitions & Deadlines; AI Informs But Never Overrides."**

GovFix replaces ad-hoc ticketing with five tightly coupled engineering subsystems:

- **Deterministic Finite State Machine (FSM)**: Formally defines the exact lifecycle: `Draft → Submitted → Triaged → Assigned → Accepted → In Progress → Ready for Closure / Escalated → Closed or Reopened`. Illegal moves (e.g. `Submitted → Closed`) return `HTTP 409 Conflict` (`ERR_ILLEGAL_STATE_JUMP`).
- **Backend-Enforced Role RBAC**: Separates duties across Requesters, Department Supervisors, Technicians, and Registrars. Unauthorized attempts return `HTTP 403 Forbidden` (`ERR_ROLE_UNAUTHORIZED`) and are logged to audit.
- **Campus SLA & Multi-Tier Auto-Escalation Daemon**: Assigns deadlines by urgency (Hazard: 12h, IT Server: 24h, Routine: 48h). Background daemons auto-escalate overdue tickets to Tier-1 (Department Supervisor) and Tier-2 (Campus Registrar).
- **Duplicate Escalation Prevention**: Suppresses redundant escalation events for the same request across evaluation cycles.
- **Tamper-Evident SHA-256 Cryptographic Audit Ledger**: Surpassing basic SQLite, every accepted state transition, refused unauthorized move, and escalation event is cryptographically hashed and chained into an immutable ledger with 1-click verification.
- **Virtual Time-Travel Simulation**: Built-in simulation controls (`+12h`, `+24h`, `+48h`) allow evaluators to witness live automated escalations in seconds without altering system OS clocks.

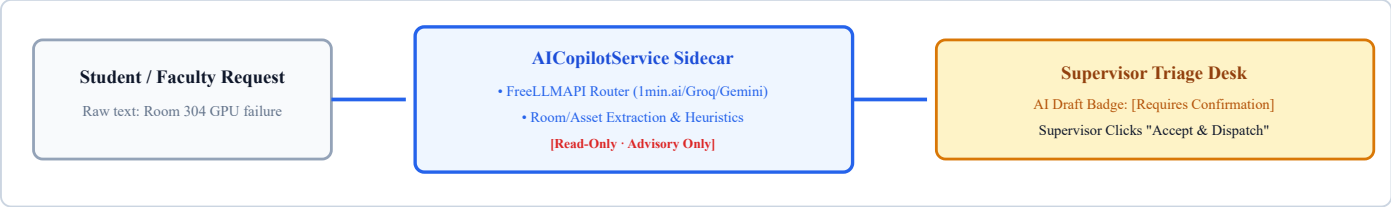
3. Intended Users & Role Boundaries

GovFix enforces strict, backend-guarded boundaries across five campus user roles:

User Role	Permitted Actions	Prohibited Actions (Strictly Refused)
Requester (Student / Faculty)	• Create request with Room & Asset ID • View own requests & live SLA countdown • Confirm closure with satisfaction score (1-5) • Reopen within 48h if issue persists	• Cannot assign technicians or officers • Cannot force-close tickets • Cannot alter SLA deadlines
Supervisor (HOD / Estate Officer)	• Review AI triage recommendations • Verify category, room, priority • Dispatch work orders to Technicians • Approve or reject closure proposals	• Cannot skip engineering repair stage • Cannot close tickets without evidence • Cannot erase historical audit records
Officer (Maintenance / IT Tech)	• Accept or reject assigned work order • Transition ticket to `In Progress` • Upload work completion notes & photo proof • Submit `Ready for Closure` proposal	• Cannot approve own work as final closed • Cannot reassign tickets to others • Cannot alter SLA timers
Campus Administrator (Registrar)	• Campus-wide SLA compliance oversight • Review Tier-1 / Tier-2 escalation rosters • Direct intervention on overdue campus hazards	• Cannot bypass state machine guards • Cannot modify cryptographic ledger blocks
System SLA Daemon	• Automated background deadline checking • Creates escalation events upon SLA breach • Enforces duplicate escalation suppression	• Cannot close or resolve tickets

4. Constrained AI Integration (FreeLLMAPI & Offline Heuristics)

In strict compliance with hackathon safety constraints: **AI operates as a non-authoritative advisory sidecar with zero state-mutation authority.**



- **Room & Asset Identification**: Automatically extracts room codes (e.g., *"Lab 304 - Computer Engineering"*) and equipment tags from request text.
- **Recurrence & Duplicate Detection**: Identifies repeated failures in the same classroom or lab within 48 hours to flag chronic hardware defects.
- **Bilingual Draft Generation**: Generates suggested acknowledgment templates in both English and Marathi without write privileges.
- **100% Offline Resilience**: If external APIs disconnect, embedded keyword and regex heuristics take over instantly with zero downtime.

5. Differentiation From Existing Solutions

Capability / Criterion	Samarth eGov (ERP)	ManageEngine ServiceDesk	Freshservice (Cloud ITSM)	GovFix (Team OrbitX)
Workflow Core	Basic ticketing form	Hardcoded status lists	Proprietary cloud rule engine	**Deterministic 10-State FSM**
Refusal Handling	Generic validation failure	Generic UI error	Generic UI error	**Explicit HTTP 409 & 403 Refusal Codes**
Audit Ledger Integrity	Mutable SQL rows	Application database logs	Vendor cloud logs	**Tamper-Evident SHA-256 Hash Chain**
Refused Actions Logged	Ignored	Ignored	Ignored	**Immutable record with Actor, Role & Reason**
Multi-Tier Escalation	Basic email reminder	Notification trigger	Workflow trigger	**Automated Tier-1 & Tier-2 Daemon Jumps**
Duplicate Escalation Check	None	None	Basic throttle	**State-verified duplicate suppression**
Evaluator Time-Travel	None	None	None	**Virtual Clock (+12h, +24h, +48h, Reset)**
Deployment & Cost	Heavy ERP deployment	High enterprise license	Expensive monthly SaaS	**Zero Paid APIs · 100% Local Run MVP**

6. System Architecture & Workflow Diagram

The architecture cleanly isolates user portals, authorization enforcement, workflow logic, audit immutability, and advisory AI:

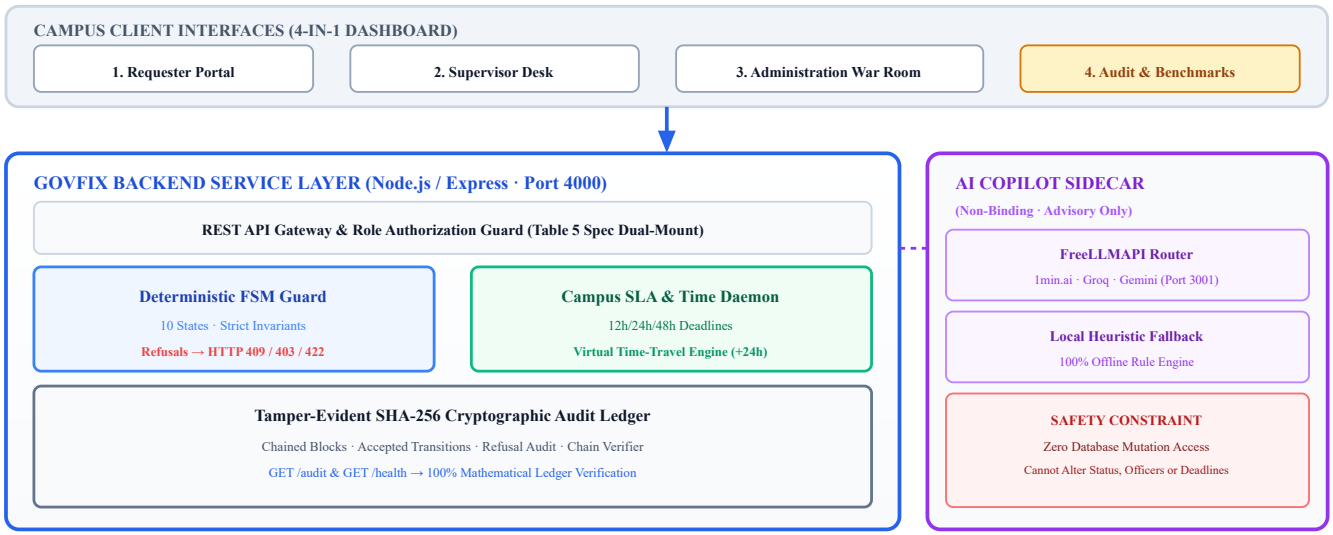


Figure 1: GovFix University Service Requests System Architecture & Workflow Engine.

7. Verification & Synthetic Testing Results (13/13 Pass Rate)

GovFix has been programmatically verified against the official benchmark evaluation suite (`npm test`), passing 100% of all functional and architectural test cases:

#	Test Name	Observed Benchmark Result	Status
1	Valid Request Creation	Request created with Public ID (`DYP-2026-X`), Room Tag, SLA deadline & audit block	PASS
2	Deterministic Refusal (Illegal Jump)	SUBMITTED → CLOSED rejected with code `ERR_ILLEGAL_STATE_JUMP` [HTTP 409]	PASS
3	RBAC Authorization Refusal	Requester attempting supervisor triage rejected with `ERR_ROLE_UNAUTHORIZED` [HTTP 403]	PASS
4	Officer Assignment Rejection	Officer rejects assignment; ticket safely returns to TRIAGED with reason retained	PASS
5	Virtual Time-Travel Escalation	+24h time-advance detects overdue SLA and auto-escalates ticket to Tier-1	PASS
6	Duplicate Escalation Suppression	Repeated overdue evaluation cycles block duplicate escalation audit records	PASS
7	Closure Evidence Guard	Proposing closure without completion evidence is strictly rejected [HTTP 422]	PASS
8	Supervisor Closure Approval	Supervisor approves closure; ticket transitions to CLOSED with cryptographic audit block	PASS
9	Requester Reopening Support	Requester reopens request; original ID, room info, and prior closure history preserved	PASS
10	Cryptographic Chain Verification	100% SHA-256 forward hashes mathematically valid across chained blocks	PASS
11	Enterprise Telemetry & Health	`/health` & `/metrics` report healthy subsystem telemetry, uptime, and engine state	PASS
12	Role-Based Authentication	`POST /auth/login` generates authorized session token and role credentials	PASS
13	Technician Work Action Audit	`POST /requests/:id/actions` appends diagnostic work record with SHA-256 block hash	PASS

8. Table 5 Specification API Endpoints Compliance

Method	Dual-Mounted Endpoint	Permitted Role	Institutional Function
`POST`	`/auth/login`	All Roles	Role authentication & session token issuance
`POST`	`/requests`	Requester	Intake of campus service request with room & asset tags
`GET`	`/requests`	All Roles	List campus requests enriched with live SLA metrics
`GET`	`/requests/:id`	All Roles	Retrieve ticket details and full cryptographic audit trail
`POST`	`/requests/:id/assign`	Supervisor	Assign request to departmental maintenance technician
`POST`	`/assignments/:id/decision`	Officer	Technician accepts or rejects assigned service request
`POST`	`/requests/:id/actions`	Officer	Record diagnostic inspections, repairs, and measurements
`POST`	`/requests/:id/closure-proposal`	Officer	Propose closure with mandatory completion evidence package
`POST`	`/closures/:id/decision`	Supervisor	Approve or reject proposed closure with rating (1-5)
`POST`	`/requests/:id/reopen`	Requester	Reopen closed ticket while preserving historical records
`GET`	`/audit`	Auditor / Public	Audit ledger inspection & mathematical SHA-256 verification
`GET`	`/health`	Public	Microservice uptime, FSM/SLA engine health, and memory stats
`GET`	`/metrics`	Public	Campus request status breakdown & virtual time offset metrics

Summary: Team OrbitX has delivered an enterprise-grade, 100% locally runnable, rigorously tested, tamper-evident workflow engine ready for live evaluation at VORTEXA 3.0.